ORION - UAV Design Hackathon: Technical Proposal

Mobile-Network Relay Drone for Disaster Zones

Team Name: Zephyr_Senior

Abstract

This document outlines the conceptual design, structural analysis, mass budget, and power allocation for our autonomous Unmanned Aerial Vehicle (UAV) developed for ORION. The system is engineered to optimize payload capacity, flight endurance, and structural rigidity under strict weight constraints.

1 Problem Statement

Restore temporary 4G or LoRa connectivity over a disaster area using a tethered or free-flying relay platform.

2 Objectives

- **Endurance:** Achieve a minimum flight time of 4hours with a 2kg communication payload.
- **Structural Integrity:** Design a robust frame optimized for maximum structural integrity, capable of withstanding the required flight maneuver limit load and wind speed of 10m/s.
- **Coverage:** Ensure connectivity over a 3km coverage radius.
- **Fail-safe Protocol:** Implement a fail-safe protocol to trigger immediate Return-to-Land (RTL) upon telemetry link disruption or critical voltage drops.

3 Mass Budget

Accurate mass tracking is critical to ensure the UAV stays within its regulatory and performance thresholds. The table below details the component mass breakdown.

Component	**Quantity**	**Unit Mass (g)**	**Total Mass (kg)**
Motor	4	277.0	1.1080
Propellers	4	80.0	0.3200
Frame	1	2000.0	2.0000
GNSS	1	10	0.0100
Flight Computer	1	73.0	0.0730
Payload	1	2000.0	2.0000
Wires	4	29.6	0.1184
Power Distribution Board	1	150.0	0.1500
ESC	4	75.0	0.3000
Battery	1	2500.0	2.5000
Buck Converter	1	100.0	0.1000
Tether	30	23.33	0.6999
Total Mass			**9.3793 kg**

Table 1: UAV Mass Budget Breakdown

4 Power Budget

Component	**Voltage (V)**	**Hover Current (A)**	**Hover Power (W)**
Motor (4x combined)	49	37	1813
Flight Controller	4.1	2.5	10.25
GPS & Telemetry	3.3	0.04	0.2
Total System	–	–	**1823.45 W**

Table 2: Electrical Power Budget

$$\text{Hover Endurance (Hours)} = \frac{\text{Battery Capacity (Wh)}}{\text{Hover Power (W)}} \times \text{Discharge Limit (80\%)}$$

Since, we are using a tethered-based system, Hover Endurance does not apply. Instead, a back-up battery present on the drone will be autonomously switched to in-case of abrupt power cutoff.

5 Conclusion

The proposed UAV design meets all fundamental criteria for the hackathon challenge. Through the integration of high-strength components and a proper power budget, our team is positioned to achieve high operational reliability and mission success.